

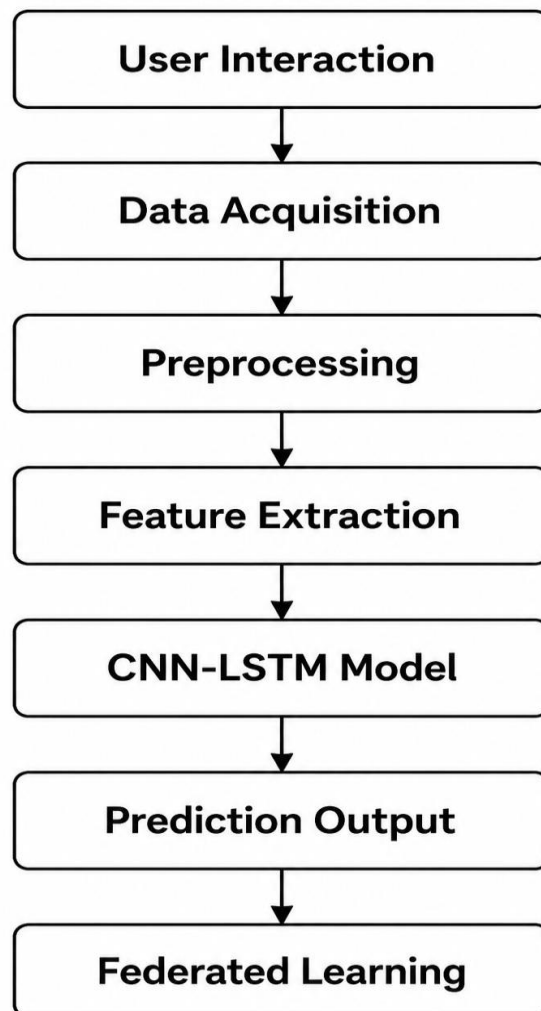
Bio-Resonant Edge-Sensing Framework

1. Introduction

This document presents an advanced detailed design of the proposed system. The design focuses on modular architecture, data flow, algorithmic processing, and system scalability. The system is positioned as an exploratory AI-based health monitoring framework.

2. Overall System Architecture

Architecture Diagram:



3. Data Flow Diagram

Level-1 Data Flow:

User → Signal Capture → Cleaning → Feature Engineering → Model → Output → Feedback

Level-2 Data Flow:

Touch Input → Capacitive Values → Noise Filtering → Normalization → Statistical Features → Temporal Sequences → CNN → LSTM → Risk Score

4. Module Design

Data Acquisition:

Captures raw touch signals such as pressure, duration, and capacitance.

Preprocessing:

Applies smoothing, normalization, and noise reduction techniques.

Feature Extraction:

Includes mean, standard deviation, FFT-based frequency features, and time-series patterns.

Model Layer:

CNN extracts spatial features; LSTM handles temporal dependencies.

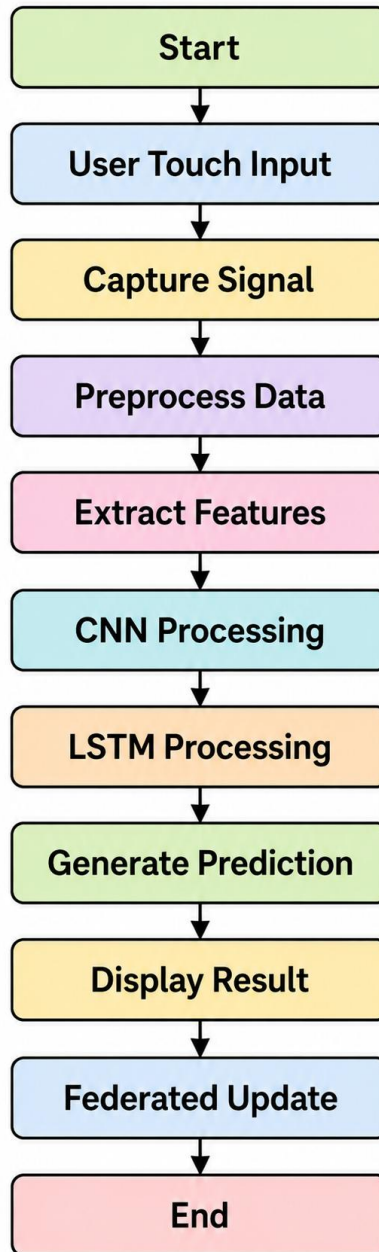
Federated Learning:

Local models are updated and aggregated without sharing raw data.

Output Layer:

Displays a risk score and trend visualization.

5. Flowchart Diagram



6. Algorithm Design

Algorithm Steps:

1. Initialize system
2. Capture signal data
3. Normalize data
4. Extract features
5. Feed into CNN layers
6. Pass to LSTM layers
7. Compute prediction score
8. Update model using federated learning
9. Output result

7. Mathematical Considerations

Signal Processing:

Normalization: $X' = (X - \text{mean}) / \text{std}$

Feature Extraction:

Mean, variance, and frequency domain features are computed.

Model:

CNN + LSTM hybrid architecture for pattern recognition.

8. Design Constraints

- Device variability
- Limited sensor precision
- No clinical validation
- Dependency on simulated data

9. Scalability and Performance

The system is scalable due to edge processing and federated learning.

Performance depends on model optimization and device capability.

10. Conclusion

This design provides a structured and modular approach. While innovative, it requires further validation for real-world deployment.